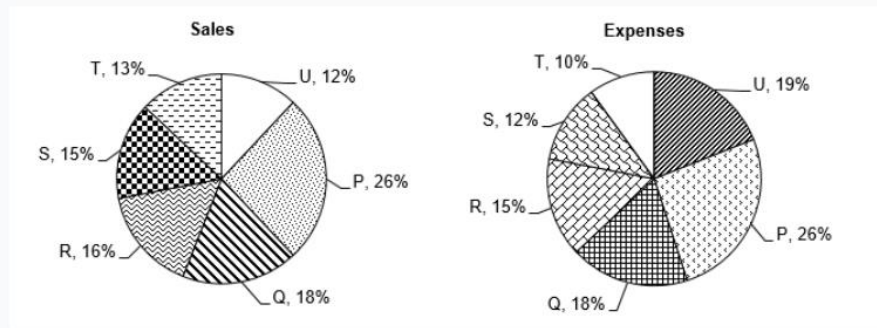


DILR PAPER-1

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The pie-charts give the breakup of sales and expenses of a company XYZ which manufactures and sells six products – P, Q, R, S, T and U.



Profit = Sales – Expenses

$$\text{Profitability (\%)} = \frac{\text{Profit}}{\text{Expenses}} \times 100$$

Solution:

Let the total expenses be ₹100 and the total sales be ₹120.

Expenses on product R = ₹15

Sales of product R = ₹19.2

$$\text{The profit percentage} = \frac{4.2}{15} \times 100 = 28\%.$$

Choice (B)

If the company made a profit, at least how many of the six products made a profit?

☐ a) 5

☐ b) 4

☒ c) 3 (Your answer - ✖)

☐ d) 2

Solution:

If the company made a profit, at least products P, Q, R, S and T would have made a profit.

Choice (A)

If all the six products made a profit, what was the least profitability of company XYZ?

- ☐ a) 51.52%
- ☐ b) 56.25%
- ☐ c) 58.33%
- ☐ d) 61.25%

Solution:

The overall profitability would be the least when product U made a very small profit. If the total sales is 'X' and the total expenses is 'Y',
 $0.12X = 0.19Y$

$$\frac{X}{Y} = \frac{19}{12}$$

The profit percentage = $\frac{7}{12} \times 100 = 58.33\%$.

Choice (C)

If the company made a profit on only two products, what was the minimum loss (%) of the company?

- ☐ a) 5.0
- ☐ b) 6.25
- ☐ c) 7.5
- ☐ d) 9.3

Solution:

The two products on which the company would have made a profit are S and T. The overall loss would be the lowest when the company had no profit or loss on the third product (R). If the total sales are X and total expenses are Y, $0.16X = 0.15Y$

$$\frac{X}{Y} = \frac{15}{16}$$

$$\text{Loss (\%)} = \frac{Y-X}{Y} \times 100 = \frac{16-15}{16} \times 100 = \frac{1}{16} \times 100 = 6.25\%$$

Choice (B)

If not all products made a profit, what can be the maximum profitability of any product?

- ☐ a) 40
- ☐ b) 50
- ☐ c) 62.5
- ☐ d) None of these

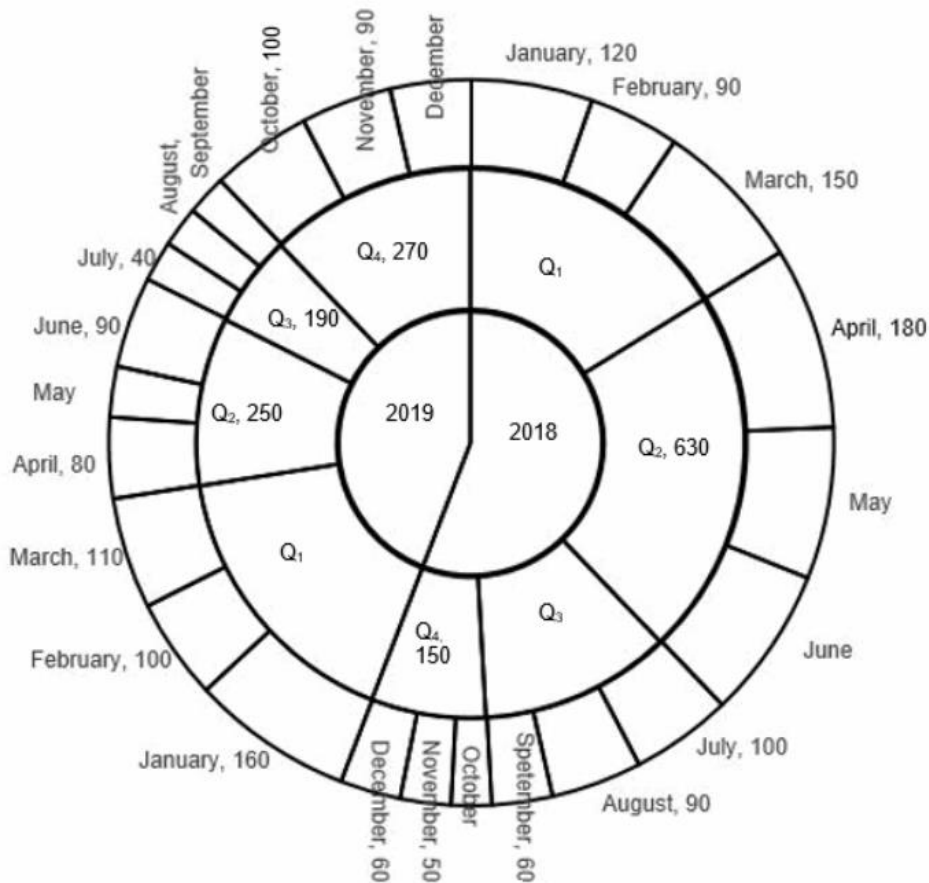
Solution:

It is similar to the previous question when we calculated the minimum profitability when all the six products made a profit. (when sales of U is just less than the expenses).
 $0.12X = 0.19Y$

$$\frac{X}{Y} = \frac{19}{12}$$

The highest profitability (for T) = 105.83%

Choice (D)



What is the percentage change in sales in December 2019 as compared to the sales in December 2018?

- ☐ a) 20
- ☐ b) 25
- ☐ c) 33.33
- ☐ d) 40

Solution:

The sales in the different months for 2018 and 2019 are as follows:

	2018	2019
Jan	120	160
Feb	90	100
Mar	150	110
Apr	180	80
May	210	80
June	240	90
July	100	40
Aug	90	60
Sept	60	90
Oct	40	100
Nov	50	90
Dec	60	80

The percentage change in sales is $= \frac{20}{60} \times 100 = 33.33\%$

Choice (C)

In which quarter of 2019 was the percentage change in sales from the same quarter of 2018 the highest?

- ☐ a) Q₁
- ☐ b) Q₂
- ☐ c) Q₃
- ☐ d) Q₄

The sales in the different quarters are

	2018	2019
Q ₁	360	370
Q ₂	630	250
Q ₃	250	190
Q ₄	150	270

The percentage change was the highest in Q₄.

Choice (D)

During which quarter is the percentage change in sales from the previous quarter's sales the highest?

- ☐ a) Q₁, 2018
- ☐ b) Q₂, 2018
- ☐ c) Q₁, 2019
- ☐ d) Q₄, 2019

Solution:

The sales in the different months for 2018 and 2019 are as follows:

	2018	2019
Jan	120	160
Feb	90	100
Mar	150	110
Apr	180	80
May	210	80
June	240	90
July	100	40
Aug	90	60
Sept	60	90
Oct	40	100
Nov	50	90
Dec	60	80

The highest percentage increase in sales from one quarter to the next was from Q₄ of 2018 to Q₁ of 2019.

Choice (C)

During which month was the percentage increase in sales from the previous month's sales the highest?

- ☐ a) March, 2018
- ☐ b) August, 2019
- ☐ c) September, 2019
- ☐ d) None of these

Solution:

The sales in the different months for 2018 and 2019 are as follows:

	2018	2019
Jan	120	160
Feb	90	100
Mar	150	110
Apr	180	80
May	210	80
June	240	90
July	100	40
Aug	90	60
Sept	60	90
Oct	40	100
Nov	50	90
Dec	60	80

The percentage increase in sales from one month to the next was the highest from December 2018 to January 2019.
Choice (D)

What is the percentage change in the total sales from 2018 to 2019?

- ☐ a) 22.3
- ☐ b) 20.6
- ☐ c) 18.1
- ☐ d) 24.5

Solution:

The sales in the different months for 2018 and 2019 are as follows:

	2018	2019
Jan	120	160
Feb	90	100
Mar	150	110
Apr	180	80
May	210	80
June	240	90
July	100	40
Aug	90	60
Sept	60	90
Oct	40	100
Nov	50	90
Dec	60	80

Total sales in 2018 = 1390

Total sales in 2019 = 1080

The percentage change = $\frac{310}{1390} \times 100 = 22.3$.

Choice (A)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The following table gives the details of a project which is divided into ten tasks – A through J. The tasks can be done in any order as long as other tasks which are required to be done before them is/are completed. The project is completed only when all the tasks are done.

Task	Duration (in minutes)	Other tasks to be completed before starting this task
A	10	G
B	9	D
C	13	D
D	15	
E	8	F, J
F	12	C, A
G	21	-
H	15	G
I	6	B
J	5	H

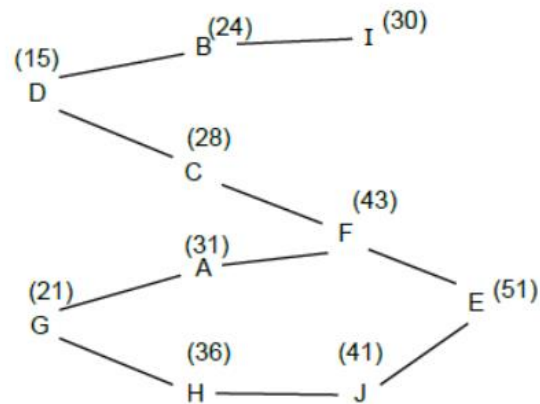
Directions for questions 11 to 13: Type in your answer in the input box provided below the question.

What is the shortest time (in minutes), from the start, in which task F can be completed?

Your Answer: 43 ✓

Solution:

The time line diagram showing the order in which the tasks are done and the shortest cumulative time taken is as given below.



Task F can be completed in 43 minutes from the start.

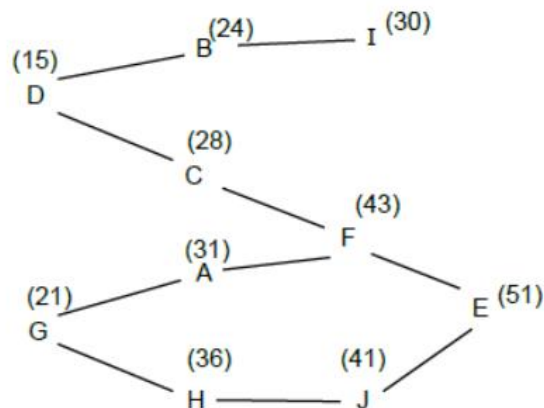
Ans: (43)

Directions for questions 11 to 13: Type in your answer in the input box provided below the question.

If the project is to be completed in the shortest possible time, what can be the minimum time gap (in minutes) between the starting of tasks B and E?

Solution:

The time line diagram showing the order in which the tasks are done and the shortest cumulative time taken is as given below.



The time gap would be minimum when both E and I get over at the same time. E has to be started 43 minutes after the start and B must start at least 36 minutes after the start. The minimum gap is seven minutes.

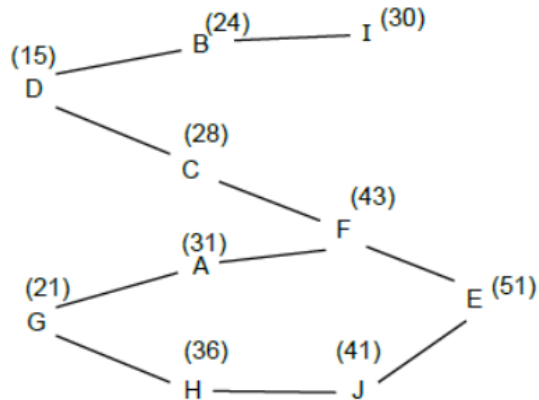
Ans: (7)

Directions for questions 11 to 13: Type in your answer in the input box provided below the question.

What is the maximum number of tasks that can be completed in 35 minutes from the start?

Solution:

The time line diagram showing the order in which the tasks are done and the shortest cumulative time taken is as given below.



The tasks that can be completed in 35 minutes from the start are D, B, I, C, G and A.
Ans: (6)

Directions for questions 14 and 15: Select the correct alternative from the given choices.
If the project is to be completed in the shortest possible time and the first task starts at 9 A.M., what can be the latest time at which task B can be started?

☐ a) 9:32 A.M.

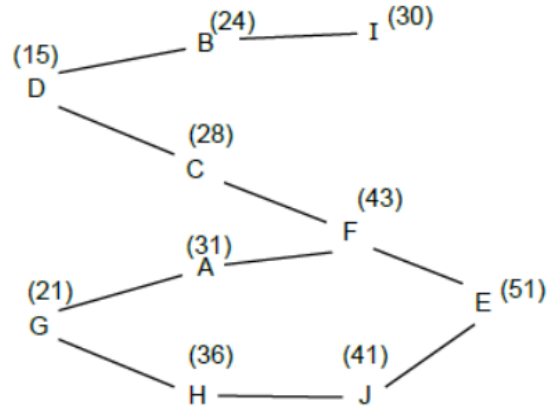
☐ b) 9:36 A.M. (Correct Answer - ✓)

☒ c) 9:38 A.M. (Your answer - ✗)

☐ d) 9:41 A.M.

Solution:

The time line diagram showing the order in which the tasks are done and the shortest cumulative time taken is as given below.



The project would be completed at 9:51 A.M. As task I can be done only after task B is completed, task B must start by at least 9:36 A.M. Choice (B)

Directions for questions 14 and 15: Select the correct alternative from the given choices.

What is the minimum time taken (in minutes) to complete the project?

☐ a) 46

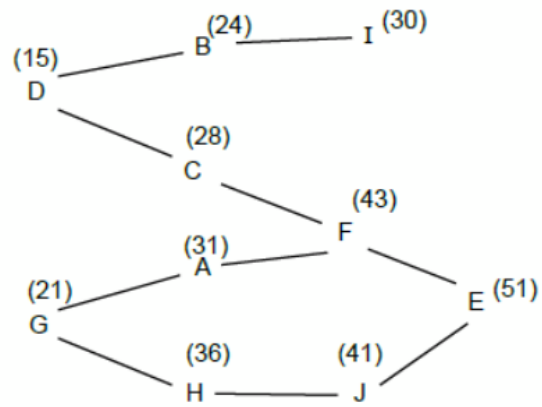
☐ b) 48

☒ c) 51 (Your answer - ✓)

☐ d) None of these

Solution:

The time line diagram showing the order in which the tasks are done and the shortest cumulative time taken is as given below.



The minimum time taken to complete the project is 51 minutes.

Choice (C)